PRE-ALGEBRA (7TH) · UNIT 13: NONLINEAR FUNCTIONS AND POLYNOMIALS

Rabbit Island Population

Model a growing rabbit population with sequences, exponential functions, and polynomials.

CP1

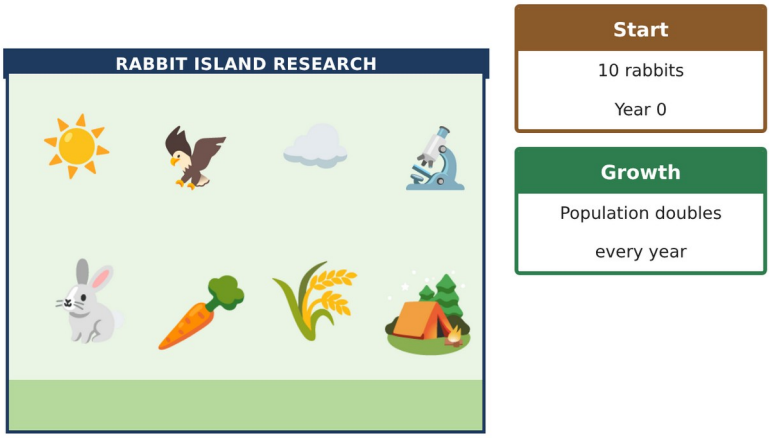
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Ten rabbits were released on a small island with no predators. The population is growing fast, the grass is disappearing, and the wildlife office needs a plan.

You will use sequences, exponential models, and polynomials to predict the population and design habitats.



Learning Goals

- I can model growth and decay with exponential functions.
- I can graph and compare linear and nonlinear functions.
- I can add, subtract, and multiply polynomials.

Project Timeline

When	What you will do
Day 1	Growth Models
Day 2	Compare Functions
Day 3	Habitat Design
Final	Wildlife Management Report + reflection

Part 1: Growth Models

Day 1 · Use the Population Survey.

Population Survey

Year 0	Year 1	Year 2	Year 3	Year 4
10 rabbits	20 rabbits	40 rabbits	80 rabbits	160 rabbits

1. Write the population as a function P(n) and find P(7).

2. Each rabbit eats 0.1 grass patch per year. Make a table for years 4–8 comparing the grass patches (200 − 15n) with the grass the rabbits need (0.1 × P(n)). In which year do the rabbits first need more grass than the island has?

3. Biologists bring foxes, and the rabbits now decrease by 30% per year from 640. Write a decay model and find the population after 3 years.

4. Is the decay model linear? Explain using the differences in your table.

Part 2: Compare Functions

Day 2 · Graph on the same axes.

5. Graph $y = 10 \cdot 2^x$ and $y = 50x + 10$ for $x = 0$ to 5. Where does the exponential pass the linear?

6. Graph $y = x^2$ for the area (sq m) of a square pen with side x. What shape is the graph?

7. Which grows fastest eventually: $50x + 10$, x^2, or $10 \cdot 2^x$?

8. Write the type of each sequence: 3, 9, 27, …; 5, 12, 19, …; 1, 4, 9, 16, …

Part 3: Habitat Design

Day 3 · Pens are rectangles with length $2x + 3$ and width $x + 1$ meters.

9. Write and simplify the area as a polynomial.

10. A larger pen is $(3x + 2)(x + 4)$. Find the difference in area between the larger pen and the first pen.

11. If $x = 6$, find both areas and the total fence for both pens.

Final Product: Wildlife Management Report

Your final product must include:

- Growth and decay models with predictions
- Graph comparing linear, quadratic, and exponential
- Sequence classification
- Polynomial habitat designs
- Management report with a recommendation

Reflection

R1. How can you tell from a table whether a function is exponential?

R2. How is multiplying binomials like finding the area of a rectangle?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Sequences, exponential models, graphs, and polynomials are correct and used to make a management plan. Goes beyond the minimum with a creative, realistic choice.	Sequences, exponential models, graphs, and polynomials are correct and used to make a management plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes tables, graphs, models, habitat designs, and a recommendation. Polished, organized, and visually engaging.	Report includes tables, graphs, models, habitat designs, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.